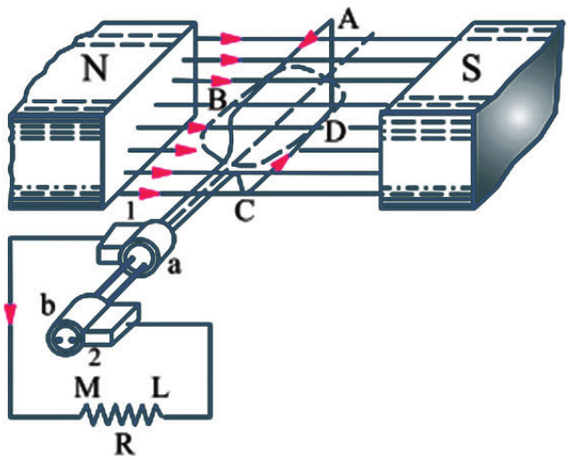



**Fig 5.2** *(a)*

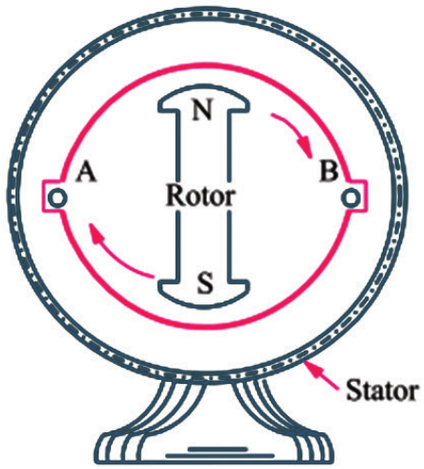



**Fig 5.2** *(b)*

In fig 5.2 (a) A stationary magnetic field and rotating coil

In fig 5.2 (b) A stationary coil and magnetic field rotating

If a coil rotates in the magnetic field or rotates inside the coil, there is an alternating emf induced in the coil. The generated alternating emf depends upon the number of turns of coils, magnetic field and the angle between the coil and magnetic field.

Induced emf e = BlV sinθ

Where

B = flux density in weber/m$^2$

l = length of the conductor in meter

V = velocity of the conductor in meter/second

Ø = angle between magnetic field and conductor

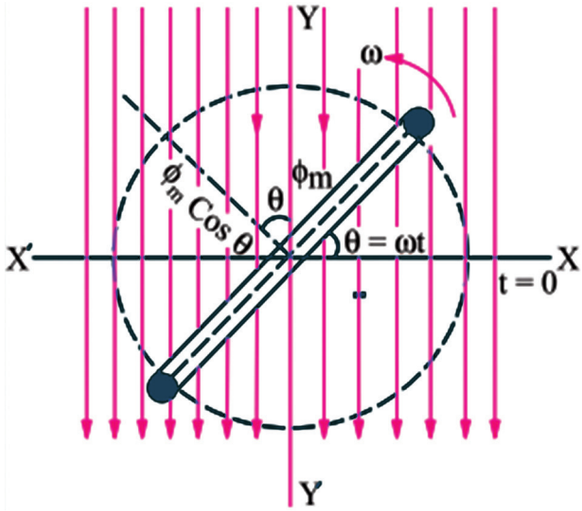



**Fig 5.3** *Rotating Coil in a Magnetic Field*

In the figure above, a rectangular coil having 'N' turns and rotating in counter clock wise direction in a uniform magnetic field with an angular velocity of 'ω' radians/sec is shown.

So, the generated A.C e.m.f. value is also depends upon the value of the angle between the magnetic field and the coil.

The sine wave may be drawn by taking the Electro motive force in 'Y' axis and time in 'X' axis.

In figure 5.4, a coil is rotating in a magnetic field in anti-clock wise direction. Let us assume that the coil is in the position 'O'. Now the angle between the magnetic field and coil is zero. Then, the e.m.f in the coil is also zero (ie, sin Ø = 0)



EMA_English_Chp05.indd 73 11/9/2020 3:09:08 PM